

Beyond Pontryagin-Optimal Quantum Control

Optimal control, robustness, and learning-based protocols

Centre for Quantum Control — working notes

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Optimal control theory · Reinforcement learning · Many-body physics

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The quantum control problem

Setting: a controlled Schrödinger equation

We steer a quantum system with time-dependent control fields $u_k(t)$:

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \left(\hat{H}_0 + \sum_k u_k(t) \hat{H}_k \right) |\psi(t)\rangle. \quad (1)$$

- $\hat{H}_0$ — the **drift** Hamiltonian (always present).
- $\hat{H}_k$ — the **control** Hamiltonians (knobs we can turn).
- $u_k(t)$ — the control **pulses** we are free to design.

Setting $\hbar = 1$ and $\hat{H}(u) \equiv \hat{H}_0 + \sum_k u_k \hat{H}_k$, the dynamics are

$$\frac{d}{dt} |\psi(t)\rangle = -i \hat{H}(u) |\psi(t)\rangle, \quad |\psi(0)\rangle = |\psi_0\rangle. \quad (2)$$

The map $u_k(\cdot) \mapsto |\psi(T)\rangle$ is what we optimise.

What do we optimise?

A control task fixes an objective functional $J[u]$. Typical choices:

Task	Objective
State transfer	$F = \langle \psi_{\text{tgt}} \psi(T) \rangle ^2$
Gate synthesis	$F = \frac{1}{d^2} \left \text{Tr}(U_{\text{tgt}}^\dagger U(T)) \right ^2$
Entanglement	maximise an entanglement monotone of $ \psi(T)\rangle$
Sensing	maximise quantum Fisher information $F_Q(T)$
Energy/time	minimise $\int_0^T \sum_k u_k^2 dt$ or minimise T

All share the structure

$$\max_{u(\cdot)} J[u] = \Phi(|\psi(T)\rangle) - \int_0^T \ell(\psi(t), u(t)) dt,$$

a terminal reward Φ plus a running cost ℓ .

Pontryagin's Maximum Principle

Pontryagin's Maximum Principle (PMP)

PMP gives **necessary conditions** for an optimum of a constrained dynamical optimisation. Classically, for $\dot{x} = f(x, u)$ one introduces a costate $p(t)$ and the **control (Pontryagin) function**

$$\mathcal{H}(x, p, u) = p^\top f(x, u) - \ell(x, u).$$

The optimal triple (x^*, p^*, u^*) satisfies

$$\dot{x} = +\partial_p \mathcal{H}, \quad \dot{p} = -\partial_x \mathcal{H}, \quad (3)$$

$$u^*(t) = \arg \max_{u \in \mathcal{U}} \mathcal{H}(x^*, p^*, u). \quad (4)$$

Key point

The control is chosen to *maximise* $\mathcal{H}$ pointwise in time over the admissible set $\mathcal{U}$. The costate p propagates the sensitivity of the objective *backwards* from the terminal time.

Derivation I — the augmented functional

We maximise $J = \Phi(|\psi(T)\rangle)$ subject to (1). Adjoin the dynamics with a costate $|\chi(t)\rangle$ (a Lagrange multiplier that is itself a ket), using 2 Re to vary bra and ket consistently:

$$\bar{J} = \Phi(|\psi(T)\rangle) - 2 \operatorname{Re} \int_0^T \langle \chi | \dot{\psi} + i \hat{H}(u) \psi \rangle dt. \quad (5)$$

Integrate the $\langle \chi | \dot{\psi} \rangle$ term by parts:

$$-2 \operatorname{Re} \int_0^T \langle \chi | \dot{\psi} \rangle dt = -2 \operatorname{Re} \langle \chi | \psi \rangle \Big|_0^T + 2 \operatorname{Re} \int_0^T \langle \dot{\chi} | \psi \rangle dt. \quad (6)$$

Collecting terms, the first variation in $|\psi\rangle$ is

$$\delta \bar{J} = \underbrace{\delta \Phi - 2 \operatorname{Re} \langle \chi(T) | \delta \psi(T) \rangle}_{\text{boundary}} + 2 \operatorname{Re} \int_0^T \langle \dot{\chi} + i \hat{H} \chi | \delta \psi \rangle dt. \quad (7)$$

(using $\langle \chi | i \hat{H} \delta \psi \rangle = \langle -i \hat{H} \chi | \delta \psi \rangle$ since $\hat{H} = \hat{H}^\dagger$).

Derivation II — costate equation & terminal condition

Stationarity for arbitrary interior variations $\delta\psi(t)$ forces the integrand to vanish:

$$\boxed{\frac{d}{dt} |\chi(t)\rangle = -i \hat{H}(u) |\chi(t)\rangle} \quad (8)$$

The costate obeys the same Schrödinger equation as the state. The boundary term fixes the terminal condition. For state transfer, $\Phi = |\langle\psi_{\text{tgt}}|\psi(T)\rangle|^2$ gives

$$\delta\Phi = 2 \operatorname{Re} \left[\langle\psi(T)|\psi_{\text{tgt}}\rangle \langle\psi_{\text{tgt}}|\delta\psi(T)\rangle \right],$$

so matching the boundary term yields

$$|\chi(T)\rangle = |\psi_{\text{tgt}}\rangle \langle\psi_{\text{tgt}}|\psi(T)\rangle = \hat{P}_{\text{tgt}} |\psi(T)\rangle. \quad (9)$$

- Forward: integrate $|\psi\rangle$ from $|\psi_0\rangle$.
- Backward: integrate $|\chi\rangle$ from $\hat{P}_{\text{tgt}} |\psi(T)\rangle$.

Derivation III — the gradient (GRAPE)

Varying the *control* in $\bar{J}$ (with $\delta\hat{H} = \sum_k \delta u_k \hat{H}_k$):

$$\delta\bar{J} = -2 \operatorname{Re} \int_0^T \langle \chi | i \delta\hat{H} \psi \rangle dt = \int_0^T \sum_k \delta u_k(t) \underbrace{2 \operatorname{Im} \langle \chi(t) | \hat{H}_k \psi(t) \rangle}_{= \delta J / \delta u_k(t)} dt. \quad (10)$$

$$\boxed{\frac{\delta J}{\delta u_k(t)} = 2 \operatorname{Im} \langle \chi(t) | \hat{H}_k \psi(t) \rangle} \quad (11)$$

- This is exactly the **GRAPE** gradient used in numerical pulse optimisation: forward-propagate $|\psi\rangle$, backward-propagate $|\chi\rangle$, overlap with each $\hat{H}_k$.
- Gradient ascent $u_k \leftarrow u_k + \eta \delta J / \delta u_k$ climbs toward a PMP stationary point.

The structure of optimal pulses

Bang-bang controls from the switching function

The control function is **affine** in u_k :

$$\mathcal{H} = \text{Im} \left\langle \chi \left| \hat{H}_0 \right| \psi \right\rangle + \sum_k u_k \underbrace{\text{Im} \left\langle \chi \left| \hat{H}_k \right| \psi \right\rangle}_{\phi_k(t)}.$$

Maximising over the box $u_k \in [u_{\min}, u_{\max}]$ a *linear* function pushes u_k to an endpoint, governed by the **switching function** $\phi_k(t)$:

$$u_k^*(t) = \begin{cases} u_{\max}, & \phi_k(t) > 0, \\ u_{\min}, & \phi_k(t) < 0. \end{cases} \quad (12)$$

Time-optimal spin flip (single control)

A single switch at t_1 gives the textbook bang–bang pulse

$$u(t) = \begin{cases} +u_{\max}, & t < t_1, \\ -u_{\max}, & t > t_1. \end{cases}$$

What if $\phi_k(t) = 0$ on a whole interval, not just at isolated points? Then $\mathcal{H}$ does not determine u_k at first order — a **singular arc**.

- One differentiates $\phi_k(t) = 0$ repeatedly in time until u_k reappears:

$$\dot{\phi}_k = \text{Im} \left\langle \chi \left| [\hat{H}(u), \hat{H}_k] \psi \right\rangle, \quad \dots$$

using $\frac{d}{dt} \text{Im} \langle \chi | A \psi \rangle = \text{Im} \left\langle \chi \left| [A, \hat{H}(u)]^* \psi \right\rangle$ -type identities.

- The **generalised Legendre–Clebsch (Kelley) condition** $(-1)^q \partial_{u_k} \left(\frac{d^{2q}}{dt^{2q}} \partial_{u_k} \mathcal{H} \right) \geq 0$ must hold on the arc.

Takeaway. PMP outputs are typically *bang-bang*, *piecewise constant*, *singular*, or smooth analytic trajectories — and they are optimal *within the assumed model*.

Beyond “Pontryagin-optimal”

What could “beyond Pontryagin-optimal” mean?

PMP *is* the optimum — so at first sight nothing beats it. The phrase is meaningful once we remember PMP optimises a **model**, for a **single** Hamiltonian, over a **restricted** ansatz, usually **open-loop**, and only when the costates are **tractable**.

1. Optimal only for the model $\dot{x} = f(x, u)$.
2. Optimality vs. robustness.
3. Restricted ansatz spaces.
4. Open-loop vs. adaptive (feedback) control.
5. Many-body intractability.

Each gap is a place where machine learning / reinforcement learning (RL) can do better *in practice*.

1. Optimal only for the model

PMP solves $\dot{x} = f(x, u)$ exactly. The real device has

noise · calibration errors · drift · decoherence · model mismatch.

Write the true generator as a perturbation,

$$\hat{H}_{\text{true}} = \hat{H}(u) + \delta\hat{H}(t), \quad \mathcal{L}_{\text{true}} = -i[\hat{H}(u), \cdot] + \mathcal{D}[\cdot],$$

with a dissipator $\mathcal{D}$ (Lindblad). The model-optimal u^* need not be optimal for $\hat{H}_{\text{true}}$ or for the open-system dynamics.

Consequence

An RL agent trained *directly on hardware outcomes* optimises the true (unknown) objective and can outperform the model-optimal pulse — one of the most active directions in quantum control today.

2. Robustness vs. optimality — the flat-optimum picture

Hardware needs performance *averaged* over uncertain parameters θ :

$$J_{\text{rob}}[u] = \int d\theta \, p(\theta) F(T, \theta). \quad (13)$$

Expand F about the nominal θ_0 with $\text{Var}(\theta) = \sigma_\theta^2$:

$$J_{\text{rob}} \approx F(\theta_0) + \frac{1}{2} \sigma_\theta^2 \partial_\theta^2 F(\theta_0) + \dots \quad (14)$$

- The sharp PMP optimum maximises $F(\theta_0)$ but may have large *negative* curvature $\partial_\theta^2 F$ — it is **fragile**.
- Maximising J_{rob} trades a little peak fidelity for a **flat** optimum (small $|\partial_\theta^2 F|$): slightly slower / less optimal, but far more robust — usually preferable.

ML readily optimises the ensemble objective J_{rob} where analytic PMP struggles.

3. Restricted ansatz spaces

Numerical PMP/GRAPE almost always imposes constraints:

- bounded amplitudes $|u_k| \leq u_{\max}$,
- finite bandwidth $|\tilde{u}_k(\omega)| = 0$ for $|\omega| > \omega_c$,
- smoothness / slew-rate limits $|\dot{u}_k| \leq s_{\max}$.

The “optimum” is then optimal *only inside that constrained set* $\mathcal{U}_{\text{ansatz}} \subset \mathcal{U}$.

$$\max_{u \in \mathcal{U}_{\text{ansatz}}} J \leq \max_{u \in \mathcal{U}} J. \quad (15)$$

ML/RL can search richer pulse families (e.g. learned parameterisations, multi-tone or non-intuitive shapes) and discover structures *outside* the original ansatz — “beyond what was thought achievable”.

4. Open-loop vs. adaptive control

PMP is usually **open-loop**: $u(t)$ is fixed in advance. Modern experiments use mid-circuit measurements, feedback and adaptive protocols, so the control depends on measurement records m :

$$u = u(t, m), \quad m = \{\text{outcomes up to } t\}. \quad (16)$$

This is a **partially observed** control problem (a POMDP). The relevant object is a *policy* mapping the information state to actions,

$$u_t = \pi(s_t), \quad s_t = (\text{belief} / \text{measurement history}).$$

Why RL fits

Reinforcement learning is built to optimise feedback policies under partial observation, where a single open-loop PMP trajectory is not even the right object.

5. Many-body intractability

For small systems PMP is exact and elegant. For

$$N \sim 100 \quad \text{interacting qubits/spins:} \quad \dim \mathcal{H} = 2^N.$$

- The state $|\psi\rangle$ and costate $|\chi\rangle$ live in an exponentially large space.
- Propagating the costate exactly becomes impossible; the GRAPE gradient $2 \operatorname{Im} \langle \chi | \hat{H}_k | \psi \rangle$ cannot be evaluated.

ML can operate on **compressed** representations — tensor networks, neural quantum states, or model-free policies trained on measurable observables — discovering approximate strategies that are simply *computationally inaccessible* to conventional PMP.

Pontryagin and reinforcement learning

The bridge: PMP is the local face of dynamic programming

Dynamic programming defines the optimal value function

$$V(x, t) = \max_{u(\cdot)} \left[\Phi(x(T)) - \int_t^T \ell \, ds \right],$$

obeying the **Hamilton–Jacobi–Bellman (HJB)** equation

$$-\partial_t V(x, t) = \max_{u \in \mathcal{U}} \mathcal{H}(x, \partial_x V, u). \quad (17)$$

Along the optimal trajectory the PMP costate is the value gradient:

$$\boxed{p^*(t) = \partial_x V(x^*(t), t)} \quad (18)$$

- PMP $\Rightarrow$ a *single optimal trajectory* (open-loop).
- HJB / DP $\Rightarrow$ the *global feedback policy* $u = \pi(x)$.
- RL approximates the DP solution from samples — naturally handling noise, feedback and high dimension.

Reinforcement learning in one slide

Model the experiment as a Markov decision process: state s_t , action a_t (a pulse / measurement / reset), reward r_t , transition $P(s' | s, a)$. Maximise discounted return $\mathbb{E}[\sum_t \gamma^t r_t]$.

$$V^\pi(s) = \mathbb{E}_\pi \left[\sum_{t \geq 0} \gamma^t r_t \mid s_0 = s \right], \quad (19)$$

$$V^*(s) = \max_a \left[r(s, a) + \gamma \sum_{s'} P(s' | s, a) V^*(s') \right]. \quad (\text{Bellman optimality})$$

- No model of f or $\mathcal{D}$ is required — learn from data.
- The control law is a learned policy $u_t = \pi_{\vartheta}(s_t)$ with parameters ϑ (e.g. a neural network).
- For many protocols there is *no known PMP formulation* of the full problem — the agent explores a far larger protocol space.

Worked examples

Example A — quantum sensing and the Fisher bound

Estimate a field B encoded unitarily, $|\psi_B\rangle = e^{-iB\hat{G}T} |\psi_0\rangle$. The precision is bounded by the **quantum Cramér–Rao bound**

$$\text{Var}(\hat{B}) \geq \frac{1}{M F_Q}, \quad F_Q = 4(\langle \partial_B \psi | \partial_B \psi \rangle - |\langle \psi | \partial_B \psi \rangle|^2). \quad (20)$$

For unitary encoding this reduces to a generator variance:

$$F_Q = 4 T^2 \text{Var}(\hat{G}) = 4 T^2 (\langle \hat{G}^2 \rangle - \langle \hat{G} \rangle^2) \quad (21)$$

- Maximising $\text{Var}(\hat{G})$ over N probes gives Heisenberg scaling $F_Q \sim N^2$ (e.g. GHZ states), vs. the standard limit $F_Q \sim N$.
- PMP yields an optimal pulse sequence *for the assumed model*.
- With uncertain noise/decoherence, an RL agent trained on measured sensitivity can find sequences with larger *observed* F_Q — “beyond Pontryagin”.

Example B — autonomous quantum experiments

A heavily studied frontier: the agent decides

what pulse to apply · *when* to measure · *when* to reset · *when* to switch strategy.

The control law is a learned policy

$$u_t = \pi(s_t), \tag{22}$$

acting on an information state s_t (belief over the unknown device).

- There is often **no known Pontryagin formulation** for the full adaptive, partially observed problem.
- The agent explores a much larger space of *protocols*, not just pulse shapes — closing the loop between control and experiment.

A flagship question for the Centre

Flagship challenge

Can machine-learning and reinforcement-learning methods discover robust and adaptive quantum control protocols that outperform conventional Pontryagin-based strategies under realistic noise, uncertainty, and feedback constraints?

- It does **not** claim RL violates optimal control theory.
- It asks whether RL beats the best *practical* implementations of PMP on *real* devices.
- It is precise, falsifiable, and experimentally grounded.

Why it is a good flagship

The question naturally connects the Centre's pillars:

Methods

- optimal control theory (PMP / GRAPE / Krotov)
- reinforcement learning
- many-body physics
- quantum information

Platforms

- quantum sensing
- superconducting qubits
- neutral-atom processors
- quantum error correction
- quantum chemistry
- autonomous laboratories

PMP is the rigorous optimum **of a model** — giving costate equations, the GRAPE gradient, and bang-bang/singular structure.

“Beyond Pontryagin” lives in the gaps: noise, robustness, restricted ansatz, feedback, and intractability.

RL is the DP/feedback counterpart of PMP, and the natural tool for those gaps.